UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: ARTIFICIAL INTELLEGENCE

SUBMITTED BY & TO

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SECTION : A

SEMESTER : 4th

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LAB # 5a

Examples Of Each Topic Of The Lab

Example 13.1a

Example 13.1a.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.1a.py (3.13.2)

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```
# Define a function that prints a greeting message
def greet_user():
    print("Hello, welcome to the Python world!")

greet_user()
greet_user()
```

Example 13.1b

Example 13.1b.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.1b.py (3.13.2)

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```
# Define a function to draw a simple triangle using asterisks
def draw_triangle():
    print('*')
    print('**')
    print('***')


draw_triangle()
```

Example 13.1c

```
# Define a function that adds two numbers and prints the result
def add_two_numbers(a, b):
    result = a + b
    print(f"The sum of {a} and {b} is {result}")


add_two_numbers(3, 4)
add_two_numbers(10, 20)
```

Example 13.2a

Example 13.2a.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.2a.py (3.13.2)

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```
def multiple_print(phrase, count):
    print(phrase * count)

multiple_print('Hello', 3)
multiple_print('A', 5)
```

Example 13.2b

Example 13.2b.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.2b.py (3.13.2)

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```
def repeat_character(char, n):
    print(char * n)

repeat_character('*', 6)
repeat_character('#', 4)
```

Example 13.3

Example 13.3.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.3.py (3.13.2)

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```python
def add(a, b):
    return a + b


def is_even(n):
    return n % 2 == 0


def sum_and_product(a, b):
    return a + b, a * b


result = add(3, 5)
print(result)


even_check = is_even(4)
print(even_check)

sum_val, product_val = sum_and_product(3, 4)
print(f"Sum: {sum_val}, Product: {product_val}")
```

Example 13.4

Example 13.4.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.4.py (3.13.2)

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```python
def convert(t):
    return t * 9 / 5 + 32

def fancy_print(text, color='black', background='white', style='normal', justify='left'):
    print(f"Text: {text}, Color: {color}, Background: {background}, Style: {style}, Justify: {justify}")

print(convert(20))
print(convert(30) + 5)

fancy_print("Hello")
fancy_print("Hello", style="bold")
fancy_print("Hello", color="yellow", background="black", style="italic", justify="center")
```

Example 13.5

Example 13.5.py - C:\Users\Rifaq Ajmal\Desktop\5(1)\Example 13.5.py (3.13.2)

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```
time_left = 30

def reset():
    global time_left
    time_left = 0

def print_time():
    print(time_left)

def func1():
    i = 10
    print(f"Local i inside func1: {i}")

def func2():
    i = 100
    func1()
    print(f"Local i inside func2: {i}")


print_time()
reset()
print_time()

func2()


def func3(x):
    x = x + 1
    return x

def func4(L):
    L = L + [1]
    print(f"Modified List in func4: {L}")

a = 3
M = [1, 2, 3]
a = func3(a)
func4(M)
print(f"Original a: {a}")
print(f"Original M: {M}")
```

LAB # 5b

EXAMPLES

EXAMPLE 19.1

Example 19.1.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.1.py (3.13.2)

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```
s = 'Hello'
copy = s
s = s + '!!!!'
print('s is now:', s, 'Copy:', copy)


L = [1, 2, 3]
copy_list = L
L[0] = 9
print('L is now:', L, 'Copy of L:', copy_list)

L = [1, 2, 3]
copy_list = L[:]
L[0] = 9
print('L after modification:', L)
print('Copy of L remains:', copy_list)

x = 487
y = x
print(f'x and y reference the same object: {x == y}')

x = 721
print(f'x now points to: {x}')
print(f'y still points to the original: {y}')
```

EXAMPLE 19.2

```
L = [1, 2, 3]
t = (1, 2, 3)
print("First element of tuple:", t[0])
print("Second element of list:", L[1])

print("Slice of the tuple:", t[1:3])

print("Length of the tuple:", len(t))
print("Count of '1' in tuple:", t.count(1))
print("Index of '3' in tuple:", t.index(3))

t1 = tuple([1, 2, 3])
t2 = tuple('abcde')

print("Tuple from list:", t1)
print("Tuple from string:", t2)

grades = {('John', 'Ann'): 95, ('Mike', 'Tazz'): 87}
print("Grades for John and Ann:", grades[('John', 'Ann')])

a, b = 5, 10
a, b = b, a
print("a:", a, "b:", b)
```

EXAMPLE 19.3

Example 19.3.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.3.py (3.13.2)

File Edit Format Run Options Window Help

```
S = {1, 2, 3, 4, 5}
print("Set S:", S)

empty_set = set()
print("Empty Set:", empty_set)

L = [1, 4, 4, 4, 5, 1, 2, 1, 3]
unique_elements = list(set(L))
print("List with duplicates removed:", unique_elements)

char_set = set('this is a test')
print("Unique characters in the string:", char_set)

A = {1, 2, 3}
B = {3, 4}
union_set = A | B '
print("Union of A and B:", union_set)

intersection_set = A & B
print("Intersection of A and B:", intersection_set)

difference_set = A - B
print("Difference between A and B:", difference_set)


symmetric_diff_set = A ^ B
print("Symmetric Difference between A and B:", symmetric_diff_set)

print("Is 3 in A?", 3 in A)

A.add(5)
print("A after adding 5:", A)

A.remove(5)
print("A after removing 5:", A)

print("Is A a subset of B?", A.issubset(B))
print("Is A a superset of B?", A.issuperset(B))

squares = {i**2 for i in range(12)}
print("Squares set:", squares)
```

EXAMPLE 19.4

Example 19.4.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.4.py (3.13.2)

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```
char_value = ord('A')
print("Unicode value of 'A':", char_value)


char_from_value = chr(65)
print("Character for Unicode value 65:", char_from_value)

print(''.join([chr(i) for i in range(1000)]))
```

EXAMPLE 19.5

Example 19.5.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.5.py (3.13.2)

File Edit Format Run Options Window Help

```
numbers = [10, 3, 15, 7, 8]

sorted_numbers = sorted(numbers)


print("Original list:", numbers)
print("Sorted list:", sorted_numbers)
```

EXAMPLE 19.6

Example 19.6.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.6.py (3.13.2)

File Edit Format Run Options Window Help

```
age = 18
status = "Adult" if age >= 18 else "Minor"
print(status)
```

EXAMPLE 19.7

Example 19.7.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.7.py (3.13.2)

File Edit Format Run Options Window Help

```
letters = ['a', 'b', 'c', 'a', 'd', 'a', 'e']

count_a = 0

for letter in letters:
    if letter != 'a':
        continue
    count_a += 1

print(f"The letter 'a' appears {count_a} times.")
```

EXAMPLE 19.8

Example 19.8.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.8.py (3.13.2)

File Edit Format Run Options Window Help

```python
def safe_input():
    try:
        num = int(input("Enter an integer: "))
        return num
    except ValueError:
        print("Invalid input! Please enter a valid integer.")
        return safe_input()

def dynamic_code_execution():
    expr = input("Enter a mathematical expression (e.g., 3*(4+5)): ")
    try:
        result = eval(expr)
        print(f"The result of the expression is: {result}")
    except Exception as e:
        print(f"Error in evaluating the expression: {e}")

    code = input("Enter a block of Python code to execute (e.g., 'x = 5; print(x)'): ")
    try:
        exec(code)
    except Exception as e:
        print(f"Error in executing the code: {e}")

print("Welcome to the Dynamic Code Execution Demo!")

user_number = safe_input()
print(f"You entered: {user_number}")

dynamic_code_execution()
```

EXAMPLE 19.9a

Example 19.9 a.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.9 a.py (3.

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```python
#enumurate example

s = 'python'
for i, x in enumerate(s, start=1):
    print(i, x)
```

EXAMPLE 19.9b

Example 19.9 b.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.9 b.py (3.13.2)

File Edit Format Run Options Window Help

```python
#zip example

names = ['Alice', 'Bob', 'Charlie']
ages = [25, 30, 35]
zipped = zip(names, ages)
zipped_list = list(zipped)
print(zipped_list)
```

EXAMPLE 19.10a

Example 19.10 a.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.10 a.py (3.13.2

File Edit Format Run Options Window Help

```python
#shallow copy example

import copy

original = [[1, 2], [3, 4]]
shallow_copy = copy.copy(original)

# Modify the original
original[0][0] = 99

print("Original:", original)
print("Shallow copy:", shallow_copy)
```

EXAMPLE 19.10B

Example 19.10 b.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.10 b.py

File Edit Format Run Options Window Help

```python
#deep copy example

import copy

original = [[1, 2], [3, 4]]
deep_copy = copy.deepcopy(original)

# Modify the original
original[0][0] = 99

print("Original:", original)
print("Deep copy:", deep_copy)
```

EXAMPLE 19.11a

Example 19.11a.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.11a.py (3.13.2)

File Edit Format Run Options Window Help

```python
#translate example:

# Translation table: 'a' -> '1', 'b' -> '2'
translation_table = str.maketrans({'a': '1', 'b': '2'})
result = 'banana'.translate(translation_table)
print(result)
```

EXAMPLE 19.11B

Example 19.11b.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.11b.py (3.13.2)

File Edit Format Run Options Window Help

```
alphabet = 'abcdefghijklmnopqrstuvwxyz'
shuffled_alphabet = 'zyxwvutsrqponmlkjihgfedcba'
cipher = str.maketrans(alphabet, shuffled_alphabet)

message = 'hello'
encoded_message = message.translate(cipher)
print(encoded_message)
```

EXAMPLE 19.12a

Example 19.12a.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.12a

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```
# Statements on the same line:

x = 5; y = 10; print(x + y)
```

EXAMPLE 19.12b

Example 19.12b.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\Example 19.12b.py (3.13.2)

File Edit Format Run Options Window Help

```
#Calling multiple methods in a chain:

s = "hello world"
result = s.upper().replace("WORLD", "Python").split()
print(result)
```

EXAMPLE 20.1

*Example 20.1.py - C:\Users\Rifaq Ajmal\Desktop\5(2)\E

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```
#Importing Modules

from random import randint
print(randint(1, 10))
```

EXAMPLE 20.2

File Edit Format Run Options Window Help

```
#example of Dates and Times

from datetime import datetime
d = datetime.now()
print(d.strftime('%A, %B %d, %Y'))
```

EXAMPLE 20.3

File Edit Format Run Options Window Help

```
#OS Example (List Files):
import os
files = os.listdir('/path/to/directory')
for file in files:
    print(file)
```

EXAMPLE 22.1

File Edit Format Run Options Window

```
def square(x):
    return x * x

g = square
print(g(3))
```

EXAMPLE 22.2

File Edit Format Run Options Window Help

```
L = [(5, 4), (3, 2), (1, 7), (8, 1)]
L.sort(key=lambda x: x[1])
print(L)
```

EXAMPLE 22.4

File Edit Format Run Options Window Help

```
L = list(map(len, ['this', 'is', 'a', 'test']))
print(L)
```

EXAMPLE 22.5

File Edit Format Run Options Window Help

```
#Operator Example:


from operator import add
from functools import reduce
total = reduce(add, range(1, 101))
print(total)
```

EXAMPLE 22.6

File Edit Format Run Options Window Help

```
#Variable Arguments Example:

def product(*nums):
    prod = 1
    for i in nums:
        prod *= i
    return prod

print(product(3, 4))
print(product(2, 3, 4))
```